Turkish Sign Language Gloss and Lexicon Schema Design

1. Purpose and project alignment

This report reviews TİD grammar, particularly SOV-related word order and non-manual marking, and proposes an initial schema for a sign-to-gloss dictionary. It specifies a logical design rather than a deployed database.

Direction A: a recognised dataset class maps to a stable sign identifier and gloss; the language layer builds Turkish text and speech. Direction B: Turkish text is mapped to sign identifiers and a gloss sequence; reviewed clips are selected for the avatar. A gloss is a written sign label, not a complete movement description or a validated translation.

The project proposal specifies a JSON dictionary containing gloss, Turkish equivalents, domain and clip reference [7]. This design can generate that dictionary. Its additional reference and review tables support provenance and later maintenance. The proposal's 150-sign recognition and 300-sign avatar targets apply to the wider project, not the number of completed entries in this initial design.

2. Review method and evidence limits

A targeted initial review used journal/publisher pages and the METU thesis repository. Search concepts were "TİD word order", "Turkish Sign Language nonmanual questions", and "TİD negation corpus". Included sources explicitly address TİD structure; ASL-only rules and unsupported tutorials were excluded. This is not an exhaustive systematic review.

Evidence used: article abstracts/repository summaries for [1]-[5], including the publisher abstract for [3]. Full-text linguistic examples and dictionary videos were not independently annotated. Consequently no invented gloss sentence is presented as validated TİD. Design recommendations below are distinguished from reported linguistic findings.

3. Linguistic findings and design consequences

3.1 Word order

Arık reports preferences for subject-before-verb and object-before-verb orders in acceptability experiments [1]. Sevinç's thesis summary reports evidence for a verb-final analysis based on native-signers' grammatical judgments and gapping data [2]. These findings support discussing SOV-related patterns, while their methods and scope do not justify a context-free word-reordering algorithm.

Design consequence: dictionary entries have no fixed sentence position. Ordered utterance tokens are separate. Turkish inflection removal and rearrangement alone must not be described as a validated TİD translation procedure.

3.2 Focus and optional non-manual information

Karabüklü and Gürer elicited focus data from 20 signers. Their abstract reports longer manual signs under focus and that non-manuals need not accompany focused signs [3]. This finding concerns focus; it is not evidence that question or negation marking is generally optional.

Design consequence: optional lexical notes do not imply universal optionality in grammar. Utterance annotations record observed form and interpreted function separately, with evidence and review status.

3.3 Questions

Göksel and Kelepir examine the role of head articulation in both content questions and polar questions in TİD [4]. This supports preserving head-related annotation rather than encoding interrogativity only as an extra word. The consulted publisher abstract does not establish a complete motion-to-question mapping, so exact head directions are not prescribed here.

Design consequence: utterance_type distinguishes polar_question, content_question, declarative and unknown. A nonmanual_span may cover multiple sign positions. Exact marker forms and their extent require source examples or qualified review before production use.

3.4 Negation

Makaroğlu's corpus-based study reports that a clause can be negated by a non-manual element alone, challenging earlier manual-dominant classifications. It also argues that backward head tilt is syntactic rather than lexically restricted to NOT [5]. These are study-specific findings, not a claim that every backward head tilt expresses negation.

Design consequence: a negative utterance need not contain an artificially inserted NOT token. Its negative interpretation can be stored in an utterance-level span. Lexical notes must not be the only place where head movement is represented. Wordless utterances and frame-level timing are deferred beyond this token-based MVP schema.

Time expressions: the proposal anticipates time-related rules [7], but this review does not independently validate a universal time-first rule. Such rules remain part of later linguistic validation.

4. Logical types and lexical tables

Logical types: TEXT = Unicode string; INTEGER = whole number; BOOLEAN = true/false; DATE = ISO date; TIMESTAMP = timezone-aware instant. “Null? No” means required. PK = primary key; FK = foreign key. SQL dialect and physical storage remain team decisions. All identifier FKs use the same logical type as their referenced PK.

4.1 sign_entry

Field / logical type	Null? / key / constraint
sign_id / TEXT	No; PK; stable SIGN_0001 format.
gloss / TEXT	No; non-empty uppercase Turkish label; not globally unique.
definition_tr / TEXT	No; Turkish sense description, distinct from a word equivalent.
domain / TEXT	No; daily, health, pharmacy or public_service for the initial scope.
variant_code / TEXT	No; non-empty v1, v2, etc.; not an asserted regional label.
lexical_nonmanual_notes / TEXT	Yes; evidence-based notes, not utterance-wide grammar.
review_status / TEXT	No; draft, source_checked or expert_reviewed; default draft.

sign_id identifies a sign variant and sense. It never changes when a label is edited. A gloss/variant/definition combination is checked for duplicates during review; free-text definitions are not used as a reliable semantic uniqueness constraint. Deletion of referenced signs is restricted; IDs are not reused.

4.2 sign_equivalent

Field / logical type	Null? / key / constraint
sign_id / TEXT	No; FK to sign_entry.sign_id; part of PK.
equivalent_tr / TEXT	No; non-empty Turkish word/expression; part of PK.

Cardinality: one sign has zero or many equivalents in draft; source_checked and expert_reviewed signs require at least one. One Turkish equivalent can map to multiple sign IDs. UTF-8 and Unicode NFC normalization are required; preserve Turkish characters. Case-insensitive matching must handle Turkish İ/i and I/ı correctly. Matching rules are application logic, not assumed default database collation.

4.3 sign_source

Field / logical type	Null? / key / constraint
source_id / TEXT	No; PK.
sign_id / TEXT	No; FK to sign_entry.sign_id.
source_title / TEXT	No; identifiable publication or dictionary.
source_url / TEXT	Yes; absolute URL when available.
source_item_ref / TEXT	No; entry ID, page, figure, example or video locator.
checked_on / DATE	Yes; required when source evidence is checked.
rights_note / TEXT	Yes; recorded reuse terms or “not assessed”.

Cardinality: one sign has zero or many source records. A dictionary homepage alone is not specific evidence for a sign. Missing evidence is null or absent; no fabricated entry URL is inserted.

5. Mapping, animation and review tables

5.1 dataset_mapping

Field / logical type	Null? / key / constraint
dataset_name / TEXT	No; part of composite PK.
dataset_version / TEXT	No; part of composite PK; no unknown-version mapping accepted.
original_label / TEXT	No; part of composite PK; original label preserved.
sign_id / TEXT	No; part of composite PK; FK to sign_entry.sign_id.
source_id / TEXT	No; FK to sign_source.source_id; must belong to sign_id.
mapping_status / TEXT	No; candidate or confirmed; default candidate.

A dataset class may have several candidate sign IDs. Only one confirmed mapping is allowed per dataset_name/version/original_label in the MVP. Ambiguous classes remain candidates and are not used as automatic lookup results. Different dataset classes may map to the same sign. Dataset terms and sign equivalence must be checked before confirmation.

5.2 animation_clip

Field / logical type	Null? / key / constraint
clip_id / TEXT	No; PK.
sign_id / TEXT	No; FK to sign_entry.sign_id.
asset_path / TEXT	No; controlled relative asset path or approved URI; non-empty.
review_status / TEXT	No; draft, source_checked or expert_reviewed; default draft.
is_default / BOOLEAN	No; default false; at most one true per sign.

Cardinality: one sign has zero or many clips; each clip belongs to one sign. A clip marked source_checked has been compared with a specific reference; expert_reviewed additionally records qualified review. A matching label alone is not a clip review. Export selects only the default expert_reviewed clip; there is no arbitrary first-row fallback.

5.3 review_event

Field / logical type	Null? / key / constraint
review_id / TEXT	No; PK; append-only history.
sign_id / TEXT	No; FK to sign_entry.sign_id.
clip_id / TEXT	Yes; FK to animation_clip.clip_id; must belong to sign_id.
target_status / TEXT	No; draft, source_checked or expert_reviewed.
source_id / TEXT	Yes; FK to sign_source.source_id; same sign; required for promotions.
reviewer_name / TEXT	No; accountable reviewer identity, not invented.
reviewer_role / TEXT	No; source_checker, TID_consultant or native_signer.
reviewed_at / TIMESTAMP	No; actual review time.
review_note / TEXT	No; evidence, decision and limitations.

A null clip_id targets the sign; a non-null clip_id targets that clip. One sign/clip can have many review events. A promotion event and the target review_status update occur together. The latest event determines current status. No review events are claimed for the draft examples in this report.

6. Utterances, spans and integrity rules

6.1 utterance

Field / logical type	Null? / key / constraint
utterance_id / TEXT	No; PK.
text_tr / TEXT	Yes; Turkish reference text if available.
utterance_type / TEXT	No; declarative, polar_question, content_question or unknown.
validation_status / TEXT	No; illustrative or verified; default illustrative.
evidence_ref / TEXT	Yes; required for verified; source URL plus item/example locator.
reviewer_name / TEXT	Yes; required for verified.
reviewed_on / DATE	Yes; required for verified.

6.2 utterance_sign

Field / logical type	Null? / key / constraint
utterance_id / TEXT	No; FK to utterance.utterance_id; part of PK.
position / INTEGER	No; part of PK; begins at 1; positive and contiguous per utterance.
sign_id / TEXT	No; FK to sign_entry.sign_id.

One utterance has one or more tokens; a sign may occur in many utterances and may repeat within an utterance. Empty utterances are rejected in this token-based design. Runtime text, raw camera video and user conversation logs need not be stored in these research-example tables.

6.3 nonmanual_span

Field / logical type	Null? / key / constraint
span_id / TEXT	No; PK.
utterance_id / TEXT	No; FK to utterance.utterance_id.
start_position / INTEGER	No; composite FK (utterance_id, start_position) to utterance_sign.
end_position / INTEGER	No; composite FK (utterance_id, end_position) to utterance_sign.
marker_description / TEXT	No; observed form, or explicit placeholder for illustrative examples.
function_label / TEXT	No; question, negation, focus, other or unknown.
validation_status / TEXT	No; illustrative or verified.
evidence_ref / TEXT	Yes; required for verified; source and example locator.
reviewer_name / TEXT	Yes; required for verified.
reviewed_on / DATE	Yes; required for verified.

Boundary rules: endpoints are inclusive; 1 ≤ start_position ≤ end_position. Both endpoints must exist in the same utterance. Overlap is allowed for simultaneous features. Editing token order invalidates spans until they are checked or remapped. Referenced-token deletion is restricted. Sub-sign timing and frame timestamps are future extensions; token spans do not claim exact motion timing.

Structural example only: an illustrative utterance with positions 1, 2 and 3 can have a span [1,3]. [0,3], [3,2] and [1,4] are invalid. A question function attached to [1,3] without evidence remains illustrative, and its marker_description must not pretend to describe an observed TİD movement.

7. Review workflow and export contract

Transition	Required evidence and action
New → draft	Known owner creates the candidate. No review or reference is fabricated.
draft → source_checked	A specific source locator, checked date, named reviewer and review event are required. Sign needs a Turkish equivalent; clip needs comparison with the reference.
source_checked → expert_reviewed	A TİD consultant or native signer records a qualified assessment, date and specific source in a new review event. No such reviewer is assumed available.
Any state → draft	A material change to meaning, variant or clip invalidates previous approval. Record reason in a new event; dependent confirmed mappings are rechecked.

Historical review events are retained. A source_checked state establishes a documented source comparison, not expert validation. The schema represents this workflow; it does not claim that a database implementation already enforces it.

7.1 JSON dictionary export

Export one object per sign_id. turkish_equivalents is an array of sign_equivalent.equivalent_tr values; definition_tr remains separate. Build the equivalents collection before joining clips so duplicate rows do not multiply results. Sort equivalents deterministically and remove exact duplicates. Dictionary records are keyed by sign_id, not gloss alone.

clip_file is the asset_path of the default expert_reviewed clip, provided the sign is also expert_reviewed. Otherwise it is null. Zero defaults yields null; multiple defaults fail validation. Unknown values use null, not empty fabricated paths. Development exports include drafts with explicit status; production export omits signs without qualified review, a Turkish equivalent or an eligible clip. This conservative gate is a proposed team decision; an alternative must be documented rather than silently weakening it.

The following valid JSON is a development example, not production-ready sign data:

```
{
"sign_id": "SIGN_0001",
"gloss": "OKUL",
"turkish_equivalents": ["okul"],
"definition_tr": "Eğitim kurumu",
"domain": "daily",
"variant_code": "v1",
"clip_file": null,
"review_status": "draft"
}
```

Missing mappings are reported explicitly. Recognition does not guess an ambiguous dataset class. Avatar output does not substitute a random clip. The proposal’s synonym/fingerspelling fallback belongs to the translation/animation pipeline, not to an invented dictionary match [7].

8. Illustrative entries and design walkthroughs

All 12 entries below are unverified candidates: variant_code = v1 and review_status = draft. Equivalent strings illustrate Turkish lookup values. No source record, confirmed dataset mapping, review event or clip is asserted. They are not annotated training samples and do not count as verified MVP vocabulary.

ID	Gloss	Equivalent / definition_tr	Domain
SIGN_0001	OKUL	okul / Eğitim kurumu	daily
SIGN_0002	HASTANE	hastane / Sağlık kurumu	health
SIGN_0003	DOKTOR	doktor / Hekim	health
SIGN_0004	İLAÇ	ilaç / Tedavide kullanılan ilaç	pharmacy
SIGN_0005	SU	su / İçilen su	daily
SIGN_0006	YARDIM	yardım / Destek veya yardım	daily
SIGN_0007	RANDEVU	randevu / Önceden belirlenen görüşme	public_service
SIGN_0008	BUGÜN	bugün / İçinde bulunulan gün	daily
SIGN_0009	YARIN	yarın / Bir sonraki gün	daily
SIGN_0010	GİTMEK	gitmek / Bir yere gitme eylemi	daily
SIGN_0011	ECZANE	eczane / İlaç alınan yer	pharmacy
SIGN_0012	TEŞEKKÜR	teşekkür / Teşekkür ifadesi	daily

8.1 Reasoned checks (not executed software tests)

Case	Expected design behaviour
Same gloss, different variant	Assign separate sign_id values; gloss lookup can return several candidates.
Dataset class has two candidates	Do not confirm both; resolve ambiguity or keep both candidate.
Two default clips for one sign	Reject until exactly one default remains.
Draft clip is the only clip	Development clip_file is null; production record omitted.
Span exceeds available token positions	Reject the span; foreign-key/boundary validation fails.
source_checked without evidence	Reject transition; retain draft status.

These walkthroughs test the consistency of the written specification only. No SQL, application, database migration, dataset evaluation or expert review was executed for this design report.

9. Open decisions and limitations

Team decisions: JSON versus database as the source of truth; physical database technology; production review gate; reviewer availability; dataset versions and actual class mappings; regional variant policy; default avatar selection; project vocabulary and licensing. These decisions must not be presented as previously approved team choices.

Limitations: this is a targeted initial review, not a complete TİD grammar. No exact question/negation motion-generation rules or validated sentence corpus are delivered. Draft entries have not been checked against dictionary videos [6]. Sentence-level examples are structural only. Reference access does not establish redistribution permission.

10. References

[1] E. Arık, “An Experimental Approach to Word Order in Turkish Sign Language,” bilig, no. 94, pp. 75-96, 2020. Evidence consulted: article abstract.

https://dergipark.org.tr/en/pub/bilig/article/773308

[2] A. M. Sevinç, “Grammatical Relations and Word Order in Turkish Sign Language (TİD),” MSc thesis, Middle East Technical University, 2006. Evidence consulted: institutional repository summary.

https://open.metu.edu.tr/handle/11511/16407

[3] S. Karabüklü and A. Gürer, “Prosody of focus in Turkish Sign Language,” Language and Cognition, vol. 16, no. 4, pp. 1238-1271, 2024. Evidence consulted: publisher abstract.

https://doi.org/10.1017/langcog.2024.4

[4] A. Göksel and M. Kelepir, “The phonological and semantic bifurcation of the functions of an articulator: HEAD in questions in Turkish Sign Language,” Sign Language & Linguistics, vol. 16, no. 1, pp. 1-30, 2013. Evidence consulted: publisher abstract/indexed record; detailed motion mappings not extracted.

https://benjamins.com/catalog/sll.16.1.01gok

[5] B. Makaroğlu, “A Corpus-Based Typology of Negation Strategies in Turkish Sign Language,” Journal of Linguistics Research, vol. 32, no. 2, pp. 111-147, 2021. Evidence consulted: journal abstract/indexed record.

https://dad.boun.edu.tr/en/pub/article/853176

[6] Ministry of Family and Social Services, Current Turkish Sign Language Dictionary. Intended reference for subsequent candidate verification; no individual dictionary entry is asserted as checked.

https://tidsozluk.aile.gov.tr/

[7] SignBridge-TR Project Proposal, team document, October 2026. User-supplied file: SignBridge-TR_Project_Proposal-3(2).docx. Used for bidirectional pipeline, JSON dictionary and project targets.

Web references checked during preparation on 15-16 October 2026. Bibliographic claims are limited to the material described above; unavailable full-text content is not claimed as reviewed.